

Course Title: Optimization

Course Code: IE402

Credit Structure(L-T-P-Cr): 3-0-2-4

Prerequisites: Fundamental understanding of Linear Algebra, Matrix Theory.

Suggested Textbook/references:

Operations Research by HA Taha 8th/9th Edition

An introduction to optimization by E. K. P. Chong and S. H. Zak (Wiley)

Evaluation Scheme

- In Semester Exam (30%)
- End Semester Exam (50%)
- Project (20%)

The marks of all the components would be duly considered for Grading. This is tentatively the final schedule of evaluation. Any minor change will be announced well in advance.

Course Outcome:

After the completion of course:

- The students will be able to understand the application and working knowledge of optimization in selected areas.
- The students will develop an insight and understanding with respect to formulation of optimization problems and designing the optimal solutions with applications in various domain and engineering applications.
- The students will get acquainted with applications of optimization problems for some practical system model and few relevant work being carried out.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
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Lecture Plan/Contents:

1. Introduction to optimization methods [4 HOURS]
 - Introduction
 - Formulation of an LPP
 - Graphical Method to solve an LPP
 - Nature of a solution
2. Linear programming problem [10 HOURS]
 - Algebraic method
 - Simplex method
 - Revised simplex method
 - Duality
 - Dual simplex method
 - Sensitivity analysis
3. Transportation Problem [6 HOURS]
 - North-west corner Rule
 - Row-minimum method
 - Vogel's Approximation
 - u-v method
4. Game Theory [6 HOURS]
 - Introduction to Game Theory
 - Game with pure and mixed strategies
 - $2 \times n$ and $m \times 2$ Games
 - LPP Formulation of $m \times n$ Games
5. Numerical Solution of Convex Optimization [8 HOURS]
 - Gradient Descent, Steepest Descent
 - Newton's Method
 - Conjugate Direction and Conjugate Gradient Methods